

VIMALNATH ACHUTHAN

Seeking Full-time Software Engineer roles starting July 2025

Champaign, IL | 447-902-4619 | achuthanvimalnath@gmail.com | linkedin.com/in/vimalnath-achuthan

SUMMARY

Software Engineer with 5+ years of experience building scalable, cloud-native microservices using Python, C#, JavaScript/TypeScript, React, Node.js, .NET Core, Java, and SQL/NoSQL. Experienced in performance optimization, clean architecture, and user-centric design. Seeking Software Engineer roles starting July 2025.

TECHNICAL SKILLS

Languages: Python, C#, C/C++, JavaScript/TypeScript, Java, CUDA
Frameworks & Libraries: React, Angular, Node, Redux, ASP.NET Core, Flask, Express, DevExpress, OpenLayers, REST, GraphQL, Langchain
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, SQS), GCP, Docker, Kubernetes, Jenkins, Git, RabbitMQ
Databases: MongoDB, MySQL, Neo4j, Redis
Concepts: Microservices, Distributed Systems, Parallel Computing, Data Structures & Algorithms

EDUCATION

University of Illinois Urbana Champaign <i>Master of Computer Science - GPA - 3.83/4.0</i>	Champaign, IL <i>Jan 2024 – May 2025</i>
Anna University - College of Engineering, Guindy <i>Bachelor of Engineering, Computer Science - GPA - 8.08/10.0</i>	Chennai, India <i>Aug 2015 – May 2019</i>

EXPERIENCE

Graduate Research Assistant <i>Beckman Institute, UIUC</i>	Champaign, IL <i>Aug 2024 – May 2025</i>
- Engineered a fast stitching tool for high-resolution microscopic images (RGB & FP32) with multiple blending options, achieving a 120× speedup (2 hours → less than 1 minute). - Streamlined algorithm to O(N) complexity for scalable processing and integrated into the H & E virtual staining pipeline for end-to-end automation. - Designed cross-platform support and currently used by 15+ researchers in Prof. Rohit Bhargava's group. - Benchmarked against ImageJ/Fiji on identical datasets and achieved 10× faster results with similar PSNR/SSIM quality.	
Senior Software Engineer <i>Trimble Inc.</i>	Chennai, India <i>July 2019 – Dec 2023</i>
- Collaborated on the design and development of an e-commerce platform for Trimble products and subscriptions. Re-architected e-commerce platform from monolith to microservices and reduced latency by 90 seconds, enabling real-time responsiveness. - Led a cross-functional team for Nikon-Trimble SightFusion portal and modernizing a mobile-first responsive portal, resulting in a 56% increase in daily active users. - Took the initiative to design and led the development of a resource planning portal to efficiently track and manage employee bandwidth across multiple projects, saving 50+ hours/month and adopted by 50+ teams. - Extended the portal to the finance team, improving billing accuracy and supporting fiscal planning. - Built multi-part upload support for files up to 50GB in SightFusion and added retry mechanism reducing upload failures by 80% and enabled export/backup/restore for data integrity. - Enhanced interactivity with zoom, pan, and freehand/rectangle/polygon selection (via OpenLayers) and enabled direct placement and management of over 100,000 datapoints, allowing users to efficiently navigate, annotate, and interact with spatial data. - Revamped the Nikon-Trimble SightFusion account management portal by automating license and subscription workflows and enabling reuse of 1,000+ inactive accounts and expired subscriptions from internal APIs. - Developed Loadrite mobile app for quarry logistics. Integrated real-time dashboards and push notifications and adopted by 100+ operators across 20+ sites.	

PROJECTS

LangChain-Based Research Assistant | *Python3, LangChain, GPT-4, React, Neo4j*

- Developed an AI research assistant using LangChain with a modular pipeline, implemented a Wintr crawler to crawl and scrape web/PDF data, and Neo4j based structuring.
- Integrated semantic search and GPT-4 for retrieval-augmented generation and intelligent reasoning.

Evaluation of Prompt Engineering Techniques on Code Understanding and Generation | *Python3*

- Benchmarked DeepSeekCoder 6.7B on reasoning, bug detection, translation, test generation using HumanEval and improved baseline by 75%.
- Applied advanced prompt engineering techniques including input specification, test case injection, and chain-of-thought reasoning to reduce hallucination and improve accuracy.

WiseWallet: Financial Insights Web App | *Python3, React, NodeJs, GCP, MySQL, Express*

- Built a personal finance app analyzing 500K+ transactions to deliver tailored spending insights and plans

ACCOMPLISHMENTS

- Awarded the Central Product Development Hackathon Award at Trimble Global Hackathon 2023 for outstanding contributions to product development and innovation.
- Secured Runner-Up position at Sprite Extreme, a national-level hackathon organized by Abacus 2018, CEG.